

Result -> first name that starts with a and b

Rule #2: What happens when one string runs out of characters?

This is the part that trips people up. If you've compared every character and they've all matched so far, but one string still has more letters left and the other one has ended — the string that *still has more letters* is considered the bigger (greater) one. The shorter one, having "run out," is treated as smaller.

Think of it like a dictionary: "cat" comes before "catalog". Why? Because c-a-t matches, but then "cat" just... stops. "catalog" keeps going. In dictionary order, stopping earlier = comes first = smaller.

Now let's apply this to 'Carol' vs 'c'

We're checking: is 'Carol' <= 'c' ?

Since 'c' ran out first while 'Carol' still has "arol" left to go, 'c' is the shorter/smaller one, and 'Carol' is bigger.

So: 'Carol' > 'c', which means 'Carol' <= 'c' is FALSE.

Since the query requires FIRST_NAME <= 'c', and Carol fails that check, Carol gets excluded.

Same logic for any C-name:

Name	Compare to 'c'	Result
Carol	C=c, then "arol" left over	Carol > 'c' → excluded
Chris	C=c, then "hris" left over	Chris > 'c' → excluded
Cy	C=c, then "y" left over	Cy > 'c' → excluded
C (just the letter C alone)	C=c, nothing left on either side	Equal → included

Compare that to a name like Alice vs 'c' :

1. Compare 1st letter: A vs c → A comes before c in the alphabet → Alice is smaller, full stop. We don't even need to look further.

So Alice < 'c' is straightforwardly TRUE, no weirdness — because the very first letter already decided it.

The core insight

The weirdness only happens with C-names, because 'c' is the *upper boundary* of your range, and it's a bare single letter. Any real name starting with C will always have more letters after the "C", and having more letters after a matching prefix makes you "greater than" the plain letter 'c'. So instead of "C" acting like a wildcard for "anything starting with C," it acts like a literal, exact tiny string that almost nothing else can be less than or equal to.

That's the whole trick: BETWEEN 'a' AND 'c' does not mean "starts with A, B, or C." It means "is alphabetically between the exact strings a and c" — and almost every real word that starts with C is alphabetically *after* the bare letter c.